

Alex S. Huang

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EDUCATION

University of Texas at Dallas

Dallas, TX

Bachelor of Science in Electrical Engineering | GPA: 4.0/4.0

Aug. 2025 – Expected May 2029

- **Honors/Awards:** Hobson Wildenthal Honors College, Full-Ride National Merit Scholarship, Dean's List
- **Relevant Coursework:** Linear Algebra, Statistics, Electronic Devices, Digital Logic, Circuit Analysis, Java/C++

EXPERIENCE

Robotics Research Assistant

Aug. 2025 – Present

Intelligent Robotics and Vision Lab, UT Dallas

Dallas, TX

- Leveraging LLM-reasoning and VLA robot models for zero-shot generalization of assistive tasks with SO-101 arm.
- Leading RobotReplica, a multi-organization (MIT, Cornell, UPenn, UCLA, etc.) robot benchmarking project.
- Papers published/in submission at RSS, NeurIPS, ICRA, Science Robotics, IEEEExplore under Dr. Yu Xiang.

Undergraduate Summer Research Intern

Jun. 2026 – Aug. 2026

Collaborative Robotics and Intelligent Systems Institute, Oregon State University

Corvallis, OR

- Developed & tested multimodal paddlewheel mechanism for amphibious robot propulsion under Dr. Brian Do.
- Presented research poster as 1 of 14 interns in annual Robotics Research Experience for Undergrads (REU).

Intelligent Robot Dynamics & Actuators Intern

Sep. 2024 – Aug. 2025

Sun Robotics Lab, Arizona State University

Tempe, AZ

- Engineered an image segmentation and reinforcement-learning pipeline of finger-angle estimation for robotic hand teleoperation using DeepLabV3, MobileNetV2, and OpenCV in Python, reducing costs by 90%.
- Refined robotic hand with custom CAD, soldering, and servo modifications, qualified for Regeneron ISEF 2025.

Cybersecurity / SWE Intern

June 2024 – Aug. 2024

Security Engineering for Computing Lab, Arizona State University

Tempe, AZ

- Built SmishSmashing, a GPT-4 powered Python application designed to counteract smishing attacks, integrating Twilio for SMS parsing, MongoDB for storage, and AWS EC2 for deployment.
- Optimized detection-to-response processing time to under 5 seconds.

RELEVANT ACTIVITIES & LEADERSHIP

Accessible Prosthetics Initiative – UTD Chapter | Treasurer & Sponsorship Chair

Sep. 2025 – Present

- Securing & managing \$3000/yr in funding for student organization via UTD club council and industry sponsors
- Utilizing CAD, KiCAD, and uMyo to prototype robotic arm PCBs and custom EMG signal-processing devices, achieving 50%+ performance improvements for the 2025 Dallas Abilities Expo.

AI Innovation Lab – Artificial Intelligence Society UTD Chapter | Project Member

Aug. 2026 – Present

- Building products for people and working hands-on with industry-standard AI technologies.

Solid-State Circuits Society (SSCS) – IEEE UTD Chapter | Eta Kappa Nu Member

Sep. 2025 – Present

- Learned basic CMOS VLSI design principles to fabricate traffic light microcontrollers using Cadence Virtuoso.

HONORS & AWARDS

NSF Computing Research Association Undergraduate Grant (\$10,000)

Sep. 2026

NSF Robots in the Real World REU Grant Undergraduate Awardee (\$7,000)

Jun. 2026

UT Dallas Undergraduate Research Scholar Award (\$800)

Apr. 2026

Regeneron ISEF Finalist (1 out of 1334 worldwide)

May 2025

TECHNICAL SKILLS & INTERESTS

Languages: Python, C++, HTML, MATLAB, English (Native), Mandarin (Native), Spanish

Software & Tools: GitHub, API Integration, MongoDB, AWS, PyTorch, LeRobot

Hardware: Shop Tools, Soldering, Arduino/ESP32, 3D Printing, CAD (OnShape), PCB Design (KiCAD/Altium)

Interests: Robotics, PC Building, Chamber Music, National Parks, Cooking, Fantasy Football, Pickleball